Suryansh Kalchuri
Roll Number: 39 T12
AI&DS ASD&D Lab

Experiment 1

LO2: Use Git Effectively for Version Control Source Code Management and Collaborative Management
Introduction to Git and GitHub

1. Global Username Configuration
PowerShell
git config --global user.name "Suryansh Kalchuri"
Purpose:
Sets the global Git username configuration. Every future commit created on this machine will be associated with the author profile **Suryansh Kalchuri**.

2. Global Email Configuration
PowerShell
git config --global user.email "suryanshkalchuri@gmail.com"
Purpose:
Sets the global Git email address. This email will be associated with commits and helps GitHub identify the author correctly.

3. Listing Active Configurations
PowerShell
git config --list
Purpose:
Displays all active Git configuration settings and verifies that the username and email address have been configured correctly.

4. Local Repository Initialization
PowerShell
git init
Purpose:
Initializes a new Git repository in the current directory and creates the hidden .git folder required for version control.

5. Remote Repository Cloning
PowerShell
git clone https://github.com/suryansh17code/experiment-1.git
Purpose:
Downloads the remote GitHub repository to the local computer and creates a local working copy of the project.

6. Repository Status Verification
PowerShell
git status
Purpose:
Displays the current status of the repository, including modified files, staged files, untracked files, and the active branch.

7. Creating a Python File
PowerShell
New-Item main.py -ItemType File
Purpose:
Creates a new empty Python file named main.py inside the project directory.

8. Staging Project Files
PowerShell

git add .
Purpose:
Stages all newly created and modified files in the project directory so they are ready to be committed.

9. Creating the First Commit
PowerShell
git commit -m "Add initial files for experiment-1"
Purpose:
Saves the staged changes permanently in the local Git repository and creates a record in the project history.

10. Pushing Files to GitHub
PowerShell
git push -u origin main
Purpose:
Uploads the local commits to the remote GitHub repository and connects the local main branch with the remote branch.

11. Checking File Differences
PowerShell
git diff
Purpose:
Displays changes made to files that have not yet been staged or committed.

12. Viewing Available Branches
PowerShell
git branch
Purpose:
Lists all available branches in the repository. The currently active branch is marked with an asterisk (*).

13. Creating a Feature Branch
PowerShell
git branch feature-optimization
Purpose:
Creates a new branch called feature-optimization where changes can be developed independently without affecting the main branch.

14. Switching to the Feature Branch
PowerShell
git switch feature-optimization
Purpose:
Switches the active working environment from the main branch to the feature-optimization branch.

15. Merging a Branch
PowerShell
git merge feature-optimization
Purpose:
Merges the changes from the feature-optimization branch into the currently active branch.

16. Returning to the Main Branch
PowerShell
git switch main
Purpose:
Switches the working environment back to the primary main branch.

17. Deleting a Feature Branch
PowerShell

git branch -d feature-optimization
Purpose:
Safely deletes the feature branch after it is no longer required.

18. Checking Remote Repository URLs
PowerShell
git remote -v
Purpose:
Displays the remote repository URLs used for fetching and pushing changes.

19. Pulling Remote Changes
PowerShell
git pull
Purpose:
Downloads and integrates the latest changes from the remote GitHub repository into the local repository.

20. Viewing Commit History
PowerShell
git log --oneline
Purpose:
Displays a compact version of the commit history, showing commit IDs and commit messages.

21. Removing a File from Staging
PowerShell
git restore --staged main.py
Purpose:
Removes main.py from the staging area without deleting the actual file.

22. Discarding Uncommitted Changes
PowerShell
git restore main.py
Purpose:
Discards unstaged changes made to main.py and restores the file to its previously committed version.

23. Creating a Hello World Python Script
PowerShell
New-Item main.py -ItemType File -Value 'print("Hello, World!")'
Purpose:
Creates the main.py file and adds a simple Python program that prints **Hello, World!**

24. Staging the Updated Script
PowerShell
git add .
Purpose:
Stages the updated Python script so that the changes can be saved in the Git repository.

25. Committing the Hello World Script
PowerShell
git commit -m "Add simple hello world script"
Purpose:
Creates a new commit containing the Hello World Python script.

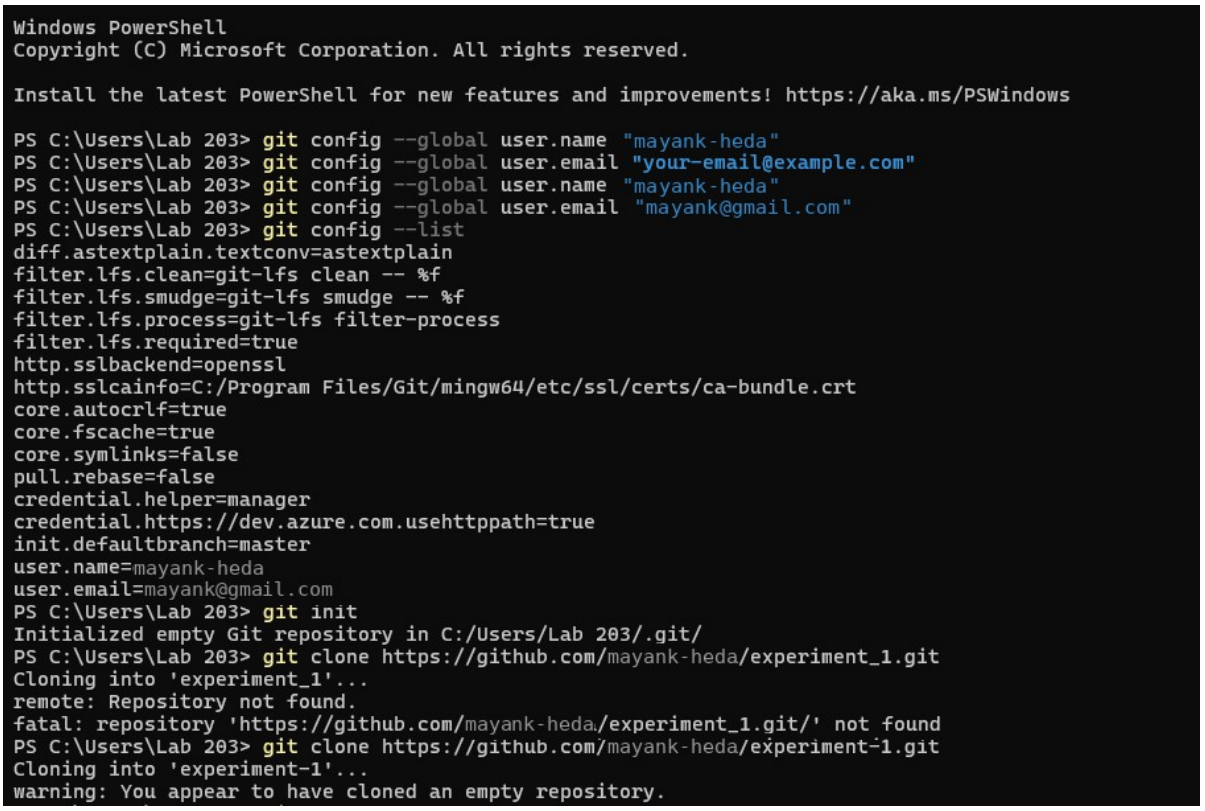
26. Final Push to GitHub
PowerShell
git push origin main
Purpose:
Uploads the latest version of the project and commit history to the GitHub repository.

Conclusion

This experiment successfully demonstrated the fundamental concepts and workflow of **Git and GitHub**. The experiment included configuring Git credentials, initializing repositories, cloning remote repositories, creating and staging files, committing changes, and pushing projects to GitHub.

Various Git operations such as checking repository status, creating and switching branches, merging branches, deleting branches, restoring files, viewing commit history, and synchronizing local and remote repositories were also performed.

Overall, this experiment provided practical knowledge of Git version control and GitHub collaboration. It demonstrated how developers can efficiently track changes, manage different versions of a project, collaborate through branches, and securely maintain project backups using remote repositories.



```
PS C:\Users\Lab 203> git status
warning: could not open directory 'Application Data/': Permission denied
warning: could not open directory 'Cookies/': Permission denied
warning: could not open directory 'Local Settings/': Permission denied
warning: could not open directory 'My Documents/': Permission denied
warning: could not open directory 'NetHood/': Permission denied
warning: could not open directory 'PrintHood/': Permission denied
warning: could not open directory 'Recent/': Permission denied
warning: could not open directory 'SendTo/': Permission denied
warning: could not open directory 'Start Menu/': Permission denied
warning: could not open directory 'Templates/': Permission denied
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .AltairRapidMiner/
        .anaconda/
        .bash_history
        .conda/
        .condarc
        .continuum/
        .dbus-keyrings/
        .docker/
        .gitconfig
        .gradle/
        .idlerc/
        .ipynb_checkpoints/
        .ipython/
        .jupyter/
        .lesshst
        .local/
        .m2/
        .matplotlib/
        .vscode/
        3D Objects/
        AppData/
        Atlassian/
        Contacts/
        Desktop/
        Documents/
        Downloads/
        Favorites/
        IntelGraphicsProfiles/
        Links/
        Music/
        NTUSER.DAT
        NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TM.blf
        NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TMContainer00000000000000000001.regtrans-ms
        NTUSER.DAT{2ad838bc-efea-11ee-a54d-000d3a94eaa1}.TMContainer00000000000000000002.regtrans-ms
        OneDrive/
        Pictures/
        Saved Games/
        Searches/
        TensorFlow/
        Untitled.ipynb
        Untitled1.ipynb
        Videos/
        anaconda3/
        anaconda_projects/
        app.py
        experiment-1/
        git-dvcs/
        git-dvcs69/
        git-t13-61/
        javac
        main.py
        myproject/
        ntuser.dat.LOG1
        ntuser.dat.LOG2
```

```
PS C:\Users\Lab 203\experiment-1> New-Item main.py -ItemType File


    Directory: C:\Users\Lab 203\experiment-1

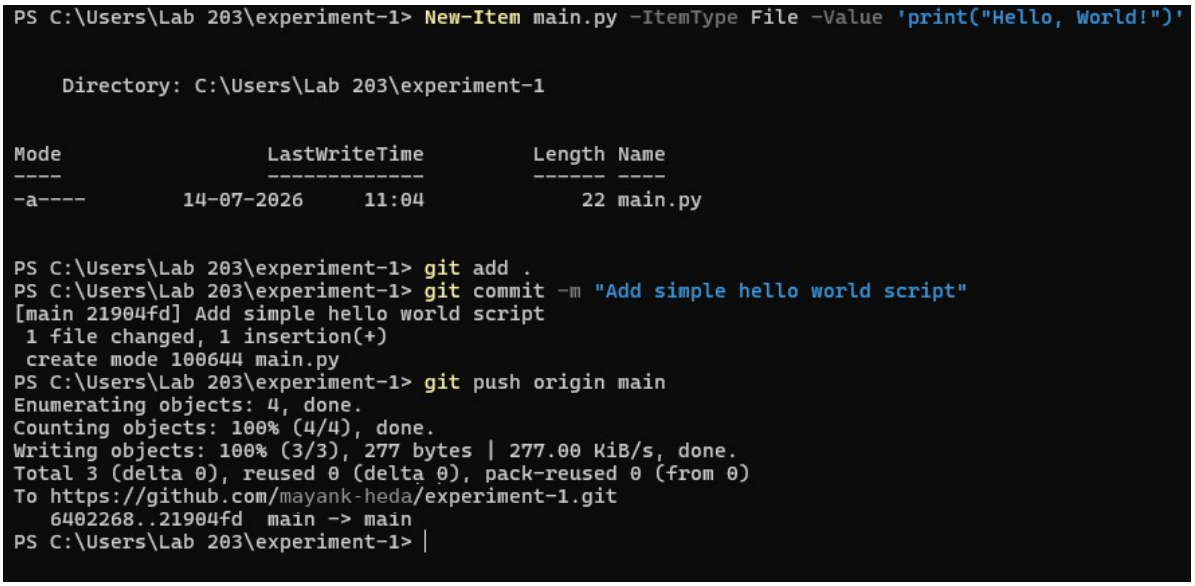

Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        14-07-2026     10:26              0 main.py


PS C:\Users\Lab 203\experiment-1> git add .
PS C:\Users\Lab 203\experiment-1> git commit -m "Add initial files for experiment-1"
[main (root-commit) 8cc79bb] Add initial files for experiment-1
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 main.py
PS C:\Users\Lab 203\experiment-1> git push -u origin main
remote: Permission to mayank-heda/experiment-1.git denied to astha9500.
fatal: unable to access 'https://github.com/mayank-heda/experiment-1.git/': The requested URL returned error: 403
PS C:\Users\Lab 203\experiment-1> git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 231 bytes | 231.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/mayank-heda/experiment-1.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\Lab 203\experiment-1> New-Item main.py -ItemType File
New-Item : The file 'C:\Users\Lab 203\experiment-1\main.py' already exists.
At line:1 char:1
+ New-Item main.py -ItemType File
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : WriteError: (C:\Users\Lab 203\experiment-1\main.py:String) [New-Item], IOException
    + FullyQualifiedErrorId : NewItemIOError,Microsoft.PowerShell.Commands.NewItemCommand

PS C:\Users\Lab 203\experiment-1> git add .
PS C:\Users\Lab 203\experiment-1> git commit -m "Add initial files for experiment-1"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
PS C:\Users\Lab 203\experiment-1> git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
```

```
PS C:\Users\Lab 203\experiment-1> git diff
PS C:\Users\Lab 203\experiment-1> git branch
* main
PS C:\Users\Lab 203\experiment-1> git branch feature-optimization
PS C:\Users\Lab 203\experiment-1> git switch feature-optimization
Switched to branch 'feature-optimization'
PS C:\Users\Lab 203\experiment-1> git checkout -b feature-optimization
fatal: a branch named 'feature-optimization' already exists
PS C:\Users\Lab 203\experiment-1> git merge feature-optimization
Already up to date.
PS C:\Users\Lab 203\experiment-1> git branch -d feature-optimization
error: cannot delete branch 'feature-optimization' used by worktree at 'C:/Users/Lab 203/experiment-1'
PS C:\Users\Lab 203\experiment-1> git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\Lab 203\experiment-1> git branch -d feature-optimization
Deleted branch feature-optimization (was 8cc79bb).
PS C:\Users\Lab 203\experiment-1> git remote add origin https://github.com/mayank-heda/experiment-1.git
error: remote origin already exists.
PS C:\Users\Lab 203\experiment-1> git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
PS C:\Users\Lab 203\experiment-1> git pull
Already up to date.
PS C:\Users\Lab 203\experiment-1> git remote -v
origin  https://github.com/mayank-heda/experiment-1.git (fetch)
origin  https://github.com/mayank-heda/experiment-1.git (push)
PS C:\Users\Lab 203\experiment-1> git log --oneline
8cc79bb (HEAD -> main, origin/main, origin/HEAD) Add initial files for experiment-1
PS C:\Users\Lab 203\experiment-1> git restore --staged code.py
error: pathspec 'code.py' did not match any file(s) known to git
PS C:\Users\Lab 203\experiment-1> git restore --staged main.py
PS C:\Users\Lab 203\experiment-1> git restore main.py
```


PS C:\Users\Lab 203\experiment-1> New-Item main.py -ItemType File -Value 'print("Hello, World!")'
Directory: C:\Users\Lab 203\experiment-1
Mode LastWriteTime Length Name
-a---- 14-07-2026 11:04 22 main.py
PS C:\Users\Lab 203\experiment-1> git add .
PS C:\Users\Lab 203\experiment-1> git commit -m "Add simple hello world script"
[main 21904fd] Add simple hello world script
1 file changed, 1 insertion(+)
create mode 100644 main.py
PS C:\Users\Lab 203\experiment-1> git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Writing objects: 100% (3/3), 277 bytes | 277.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/mayank-heda/experiment-1.git
6402268..21904fd main -> main
PS C:\Users\Lab 203\experiment-1>